

200 mm

40 mm



CAPLIPROST

Carboprost Solution for Injection USP 250 mcg/ mL

As this preparation contains benzyl alcohol, its use should be avoided in children under 2 years of age. Not to be used in neonates.

1. Name of the medicinal product

CAPLIPROST

Carboprost Solution for Injection USP 250 mcg/ mL

2. Qualitative and quantitative composition

There is following excipients used in the formulation of Carboprost Solution for Injection USP 250 mcg/ mL.

Each vial contains 332 mcg of Carboprost Tromethamine equivalent to 250 mcg Carboprost

Tromethamine USP

Benzyl Alcohol NF

Sodium Chloride USP

Sodium Hydroxide -NF

Hydrochloric Acid USP-NF

Water for Injection USP/BP

3. Pharmaceutical form

Solution for Injection

Description

A clear, colourless solution.

4. Clinical particulars

4.1 Therapeutic indications

CAPLIPROST, is indicated for aborting pregnancy between the 13th and 20th weeks of gestation as calculated from the first day of the last normal menstrual period and in the following conditions related to second trimester abortion.

- Failure of expulsion of the fetus during the course of treatment by another method;
- Premature rupture of membranes in intrauterine methods with loss of drug and insufficient or absent uterine activity;
- Requirement of a repeat intrauterine instillation of drug for expulsion of the fetus;
- Inadvertent or spontaneous rupture of membranes in the presence of a previable fetus and absence of adequate activity for expulsion.

CAPLIPROST, is indicated for the treatment of postpartum hemorrhage due to uterine atony which has not responded to conventional methods of management. Prior treatment should include the use of intravenously administered oxytocin, manipulative techniques such as uterine massage and, unless contraindicated, intramuscular ergot preparations. Studies have shown that in such cases, the use of Carboprost has resulted in satisfactory control of hemorrhage, although it is unclear whether or not ongoing or delayed effects of previously administered ecbolic agents have contributed to the outcome. In a high proportion of cases, Carboprost used in this manner has resulted in the cessation of life- threatening bleeding and the avoidance of emergency surgical intervention.

4.2 Posology and method of administration

1. Abortion and Indications 1-4:

An initial dose of 1 mL of CAPLIPROST, (containing the equivalent of 250 micrograms of Carboprost) is to be administered deep in the muscle with a tuberculin syringe. Subsequent doses of 250 micrograms should be administered at 1½ to 3½ hour intervals depending on uterine response.

An optional test dose of 100 micrograms (0.4 mL) may be administered initially. The dose may be increased to 500 micrograms (2 mL) if uterine contractility is judged to be inadequate after several doses of 250 micrograms (1 mL). The total dose administered of CAPLIPROST should not exceed 12 milligrams and continuous administration of the drug for more than two days is not recommended.

2. For Refractory Postpartum Uterine Bleeding

An initial dose of 250 micrograms of CAPLIPROST, (1 mL of Carboprost) is to be given deep, intramuscularly. In clinical trials it was found that the majority of successful cases (73%) responded to single injections. In some selected cases, however, multiple dosing at intervals of 15 to 90 minutes was carried out with successful outcome. The need for additional injections and the interval at which these should be given can be determined only by the attending physicians as dictated by the course of clinical events. The total dose of Carboprost should not exceed 2 milligrams (8 doses).

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit.

4.3 Contraindications

- Hypersensitivity (including anaphylaxis and angioedema) to CAPLIPROST, [see ADVERSE REACTIONS]
- Acute pelvic inflammatory disease.
- Patients with known active cardiac, pulmonary, renal, or hepatic disease.

CAPLIPROST, does not appear to directly affect the fetoplacental unit. Therefore, the possibility does exist that the previable fetus aborted by CAPLIPROST could exhibit transient life signs. It is not indicated if the fetus in utero has reached the stage of viability. CAPLIPROST, should not be considered a fetocidal agent.

Evidence from animal studies has suggested that certain other prostaglandins have some teratogenic potential. Although these studies do not indicate that it is teratogenic, any pregnancy termination with Carboprost that fails should be completed by some other means.

This product contains benzyl alcohol. Benzyl alcohol has been reported to be associated with a fatal “Gasping Syndrome” in premature infants.

4.4 Special warnings and precautions for use

Warnings

CAPLIPROST, (Carboprost tromethamine), like other potent oxytocic agents, should be used only with strict adherence to recommended dosages. CAPLIPROST, should be used by medically trained personnel in a hospital which can provide immediate intensive care and acute surgical facilities.

Precautions

General

Animal studies lasting several weeks at high doses have shown that prostaglandins of the E and F series can induce proliferation of bone. Such effects have also been noted in newborn infants who have received prostaglandin E1 during prolonged treatment. There is no evidence that short term administration of CAPLIPROST, can cause similar bone effects.

In patients with a history of asthma, hypo- or hypertension, cardiovascular, renal, or hepatic disease, anemia, jaundice, diabetes, or epilepsy, CAPLIPROST, should be used cautiously.

As with any oxytocic agent, CAPLIPROST, should be used with caution in patients with compromised (scarred) uteri.

Abortion

As with spontaneous abortion, a process which is sometimes incomplete, abortion induced by CAPLIPROST, may be expected to be incomplete in about 20% of cases.

Although the incidence of cervical trauma is extremely small, the cervix should always be carefully examined immediately post-abortion. Use of CAPLIPROST, is associated with transient pyrexia that may be due to its effect on hypothalamic thermoregulation. Temperature elevations exceeding 2° F (1.1° C) were observed in approximately one-eighth of the patients who received the recommended dosage regimen. In all cases, temperature returned to normal when therapy ended. Differentiation of post-abortion endometritis from drug-induced temperature elevations is difficult, but with increasing clinical experience, the distinctions become more obvious and are summarized below:

S. No.	Endometritis pyrexia	Pyrexia induced by CAPLIPROST
01	Time of onset: Typically, on third post-abortion day (38° C or higher).	Within 1 to 16 hours after the first injection.
02	Duration: Untreated pyrexia and infection continue and may give rise to other pelvic infection	Temperatures revert to pre-treatment levels after discontinuation of therapy without any other treatment.
03	Retention: Products of conception are often retained in the cervical os or uterine cavity.	Temperature elevation occurs whether or not tissue is retained.
04	Histology: Endometrium is infiltrated with lymphocytes and some areas are necrotic and hemorrhagic.	Although the endometrial stroma may be edematous and vascular, it is not inflamed
05	The uterus: Often remains boggy and soft with tenderness over the fundus, and pain on moving the cervix on bimanual examination.	Uterine involution normal and uterus is not tender.
06	Discharge: Often associated with foul-smelling lochia and leukorrhea.	Lochia normal
07	Cervical culture: The culture of pathological organisms from the cervix or uterine cavity after abortion alone does not warrant the diagnosis of septic abortion in the absence of clinical evidence of sepsis. Pathogens have been cultured soon after abortion in patients with no infections. Persistent positive culture with clear clinical signs of infections are significant in the differential diagnosis.	
08	Blood count: Leucocytosis and differential white cell counts do not distinguish between endometritis and hyperthermia caused by CAPLIPROST, since total WBC's may increase during infection and transient leucocytosis may also be drug-induced. Fluids should be forced in patients with drug-induced fever and no clinical or bacteriological evidence of intrauterine infection. Any other simple empirical measures for temperature reduction are unnecessary because all fevers induced by CAPLIPROST, have been transient or self-limiting.	

Postpartum Hemorrhage:

Increased blood pressure. In the postpartum hemorrhage series, 5/115 (4%) of patients had an increase of blood pressure reported as a side effect. The degree of hypertension was moderate and it is not certain as to whether this was in fact due to a direct effect of CAPLIPROST, or a return to a status of pregnancy associated hypertension manifest by the correction of hypovolemic shock. In any event the cases reported did not require specific therapy for the elevated blood pressure. Use in patients with chorioamnionitis. During the clinical trials with CAPLIPROST, chorioamnionitis was identified as a complication contributing to postpartum uterine atony and hemorrhage in 8/115 (7%) of cases, 3 of which failed to respond to CAPLIPROST. This complication during labor may have an inhibitory effect on the uterine response to CAPLIPROST similar to what has been reported for other oxytocic agents.

4.5 Interaction with other medicinal products and other forms of interaction

CAPLIPROST, may augment the activity of other oxytocic agents. Concomitant use with other oxytocic agents is not recommended.

Carcinogenesis, Mutagenesis, Impairment of Fertility:

Carcinogenic bioassay studies have not been conducted in animals with CAPLIPROST, due to the limited indications for use and short duration of administration. No evidence of mutagenicity was observed in the Micronucleus Test or Ames Assay.

4.6 Fertility, pregnancy and lactation

Teratogenic Effects: Pregnancy Category C

Animal studies do not indicate that CAPLIPROST, is teratogenic, however, it has been shown to be embryotoxic in rats and rabbits and any dose which produces increased uterine tone could put the embryo or fetus at risk.

Pediatric Use

Safety and effectiveness in pediatric patients have not been established.

4.7 Effects on ability to drive and use machines

No data on the effects on the ability to drive and use machines is available.

There have been reports of undesirable effects such as syncope, dizziness and somnolence which could impair the ability to drive or use machines.

Therefore, patients should refrain from driving until they know that CAPLIPROST does not affect their ability to drive or use machines.

4.8 Undesirable effects

The table below lists the adverse effects identified through clinical trials and post-marketing surveillance by System Organ Class (SOC) and frequency. Within each frequency grouping, adverse events are presented in order of decreasing seriousness. Frequencies are defined as: very common, common, uncommon, rare, very rare, or not known.

The adverse effects of CAPLIPROST are generally transient and reversible on discontinuation of therapy. The most frequent adverse reactions observed are related to its contractile effect on smooth muscles.

In the patients, some experienced vomiting and diarrhoea, some had nausea, some had a temperature increase greater than 1.1° C, and some experienced flushing.

MedDRA System Organ Class	Frequency	Undesirable Effects
<i>Infections and Infestations</i>	Uncommon	Septic shock, Urinary tract infection
	Common	Endometritis*
<i>Immune system disorders</i>	Not Known	Hypersensitivity reactions [†] (e.g. Anaphylactic reaction, Anaphylactic shock, Anaphylactoid reaction, Angioedema)
<i>Endocrine disorders</i>	Not Known	Thyrototoxic crisis [†]
<i>Psychiatric disorders</i>	Uncommon	Sleep disorder
	Not Known	Anxiety [†] , Nervousness [†]
<i>Nervous system disorders</i>	Common	Headache*
	Uncommon	Syncope vasovagal, Dizziness*, Dystonia, Paraesthesia, Somnolence, Dysgeusia, Lethargy
	Not Known	Syncope [†]
<i>Eye disorders</i>	Uncommon	Vision blurred, Eye pain
<i>Ear and labyrinth disorders</i>	Uncommon	Vertigo, Tinnitus
<i>Cardiac disorders</i>	Uncommon	Tachycardia
	Not Known	Palpitations [†]
<i>Vascular disorders</i>	Common	Flushing, Hot flush, Chills
	Uncommon	Hypertension
<i>Respiratory, thoracic and mediastinal disorders</i>	Common	Cough
	Uncommon	Asthma, Respiratory distress, Dyspnoea, Hyperventilation*, -Wheezing, Hiccups
	Not Known	Bronchospasm, Pharyngeal oedema, Choking sensation [†] , Epistaxis [†] , Dry throat [†] , Upper respiratory tract infection
<i>Gastrointestinal disorders</i>	Very common	Diarrhoea*, Nausea*, Vomiting*
	Uncommon	Haematemesis, Abdominal pain upper, Dry mouth
	Not Known	Retching [†]
<i>Skin and subcutaneous tissue disorders</i>	Uncommon	Hyperhidrosis
	Not known	Rash [†]
<i>Musculoskeletal and connective tissue disorders</i>	Uncommon	Torticollis, Back pain, Myalgia,
	Not Known	Muscle spasms, Blepharospasm [†]
<i>Reproductive system and breast disorders</i>	Common	Uterine haemorrhage, Retained placenta or membranes
	Uncommon	Uterine rupture, Uterine cervical laceration, Pelvic pain*, Breast tenderness
	Not Known	Uterine disorder
<i>General disorders and administration site conditions</i>	Uncommon	Chest discomfort, Injection site pain
	Not Known	Chest pain [†] , Asthenia [†] , Excessive thirst [†]
<i>Investigations</i>	Very common	Body temperature increased

* Events reported for both intramuscular and intra-amniotic routes of administration are marked with an asterisk. All other events were reported only for the intramuscular route.

[†] Identified from post-marketing experience

4.9 Overdose

Treatment of overdosage must be symptomatic and supportive as clinical studies with prostaglandin antagonists have not progressed to the point where recommendations may be made.

If evidence of excessive side-effects appears, the frequency of administration should be decreased or administration discontinued.

5. Pharmacological properties

5.1 Pharmacodynamic properties

CAPLIPROST administered intramuscularly stimulates in the gravid uterus myometrial contractions similar to labor contractions at the end of a full term pregnancy.

Whether or not these contractions result from a direct effect of Carboprost on the myometrium has not been determined. Nonetheless, they evacuate the products of conception from the uterus in most cases.

Postpartum, the resultant myometrial contractions provide hemostasis at the site of placentation.

CAPLIPROST also stimulates the smooth muscle of the human gastrointestinal tract. This activity may produce the vomiting or diarrhea or both that is common when CAPLIPROST is used to terminate pregnancy and for use postpartum. In laboratory animals and also in humans CAPLIPROST can elevate body temperature. With the clinical doses of CAPLIPROST used for the termination of pregnancy, and for use postpartum, some patients do experience transient temperature increases.

In laboratory animals and in humans large doses of CAPLIPROST can raise blood pressure, probably by contracting the vascular smooth muscle. With the doses of CAPLIPROST used for terminating pregnancy, this effect has not been clinically significant. In laboratory animals and also in humans CAPLIPROST can elevate body temperature. With the clinical doses of CAPLIPROST used for the termination of pregnancy, some patients do experience temperature increases. In some patients, CAPLIPROST may cause transient bronchoconstriction.

5.2 Pharmacokinetic properties

Drug plasma concentrations were determined by radioimmunoassay in peripheral blood samples collected by different investigators from 10 patients undergoing abortion. The patients had been injected intramuscularly with 250 micrograms of Carboprost at two hour intervals. Blood levels of drug peaked at an average of 2060 picograms/mL one-half hour after the first injection then declined to an average concentration of 770 picograms/mL two hours after the first injection just before the second injection. The average plasma concentration one-half hour after the second injection was slightly higher (2663 picograms/mL) than that after the first injection and decreased again to an average of 1047 picograms/mL by two hours after the second injection. Plasma samples were collected from 5 of these 10 patients following additional injections of the prostaglandin.

The average peak concentrations of drug were slightly higher following each successive injection of the prostaglandin, but always decreased to levels less than the preceding peak values by two hours after each injection.

Five women who had delivery spontaneously at term were treated immediately postpartum with a single injection of 250 micrograms of CAPLIPROST. Peripheral blood samples were collected at several times during the four hours following treatment and CAPLIPROST levels were determined by radioimmunoassay.

The highest concentration of CAPLIPROST was observed at 15 minutes in two patients (3009 and 2916 picograms/mL), at 30 minutes in two patients (3097 and 2792 picograms/mL), and at 60 minutes in one patient (2718 picograms/mL).

5.3 Preclinical safety data

Not applicable for generic products.

6. Pharmaceutical particulars

6.1 List of excipients

Tromethamine USP
Benzyl Alcohol NF
Sodium Chloride USP
Sodium Hydroxide NF
Hydrochloric Acid USP-NF
Water for Injection USP/BP

6.2 Shelf life

36 Months

6.3 Special precautions for storage

CAPLIPROST must be refrigerated at 2° to 8° C (36° to 46° F).

CAPLIPROST for single use only, discard the unused portion immediately after open.

6.4 Nature and contents of container

2 mL, 13mm clear Type I Glass Vial with 13mm grey chlorobutyl rubber stopper laminated on plug surface, Westar RS, Silicone oil level 0 and 13 mm royal blue grain finished flip top seals.
10 vials per carton.

6.5 Special precautions for disposal and other handling

CAPLIPROST is available in the following packages:

1 mL fill in 2 mL single dose vial.

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit.

CAPLIPROST to be administered deep in the muscle with a tuberculin syringe.

For Abortion and Indications 1-4: The total dose administered of CAPLIPROST should not exceed 12 milligrams and continuous administration of the drug for more than two days is not recommended.

For Refractory Postpartum Uterine Bleeding: The total dose of Carboprost should not exceed 2 milligrams (8 doses).

7. Marketing authorisation holder

Manufactured by:

Caplin Steriles Limited,

Survey No. 895 & 897,
Gummidipoondi, Tamil Nadu, 601201 India.

Code: TN/Drugs/TN00003457

Product Registration holder:

Unimed SDN. BHD.

No.53, Jalan Tembaga Sd 5/2B, Bandar Sri Damansara, 52200, Kuala Lumpur Wilayah Persekutuan Kuala Lumpur Malaysia.

8. Date of revision of the text

September - 2025



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